

Lab 9 - Report

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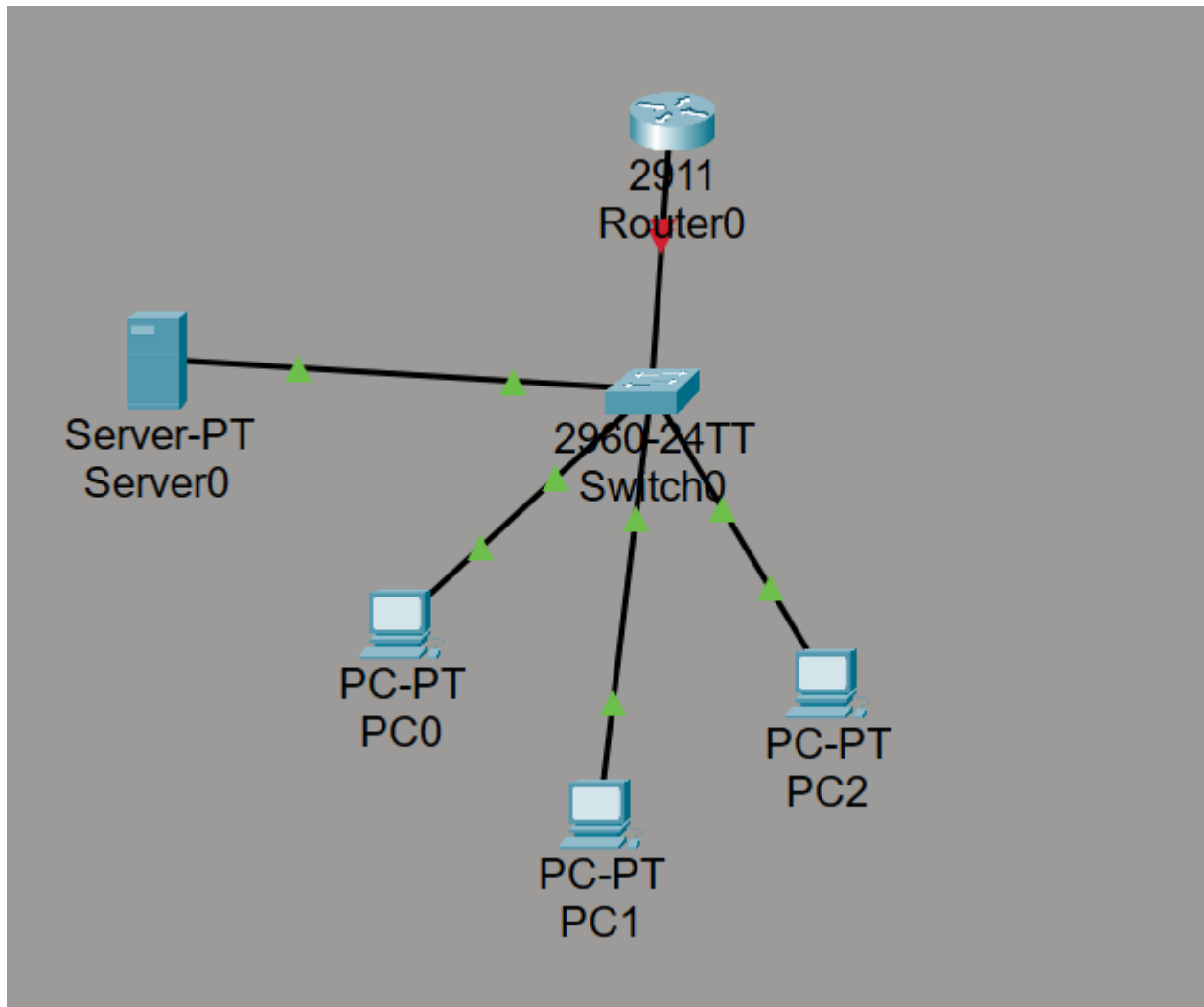
Student ID: 24B031692

Group: Mon 11:00-13:00

Configuring Network Address Translation (NAT)

Part 1: Network Setup

1.1 Topology and Addressing Scheme

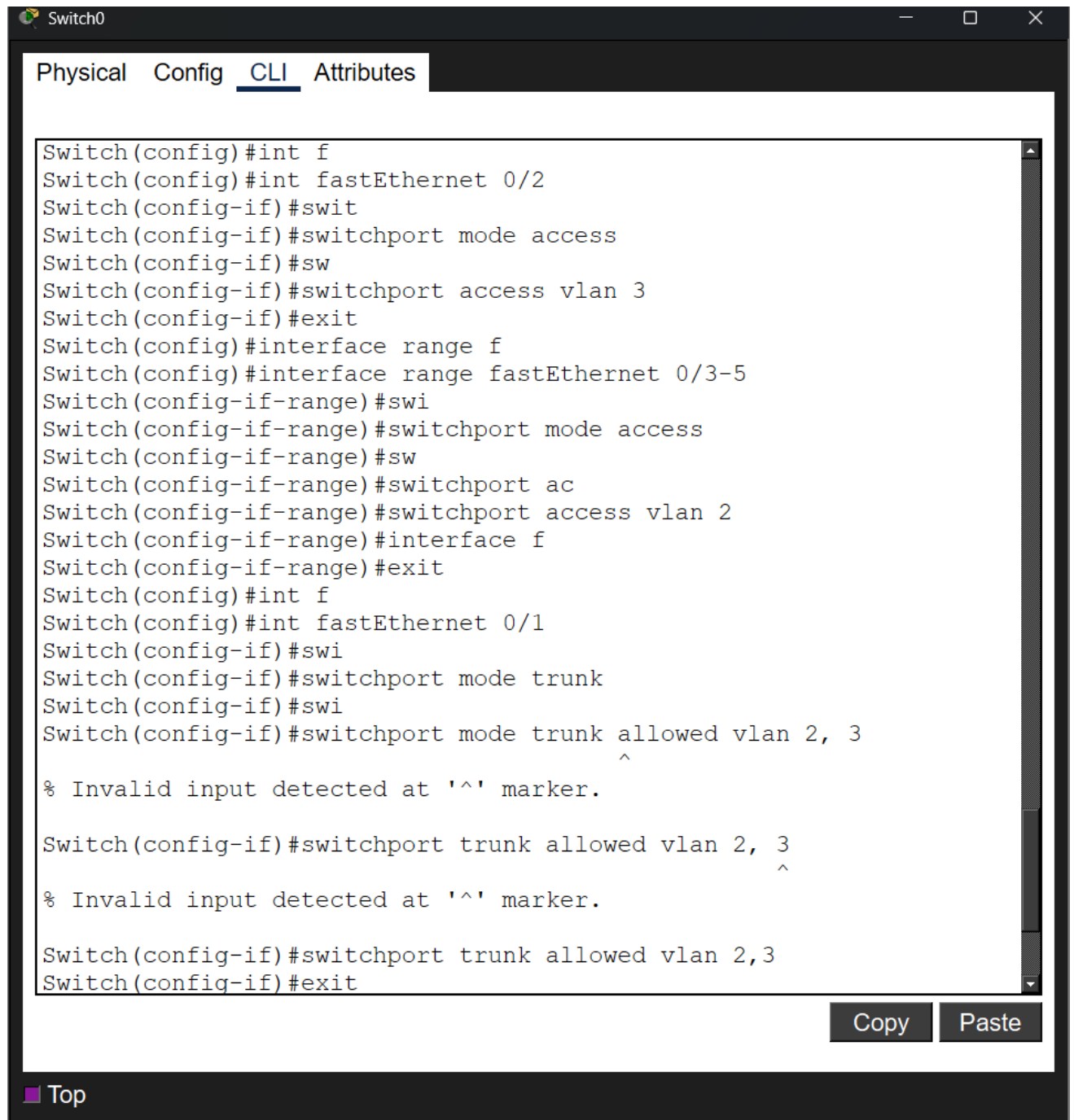


1.2 Configure VLANs on the Switch

Create VLANs

```
Switch>en
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#vlan 2
Switch(config-vlan)#name users
Switch(config-vlan)#exit
Switch(config)#vlan 3
Switch(config-vlan)#name server
Switch(config-vlan)#exit
```

Configure accesses



```
Switch0
Physical Config CLI Attributes

Switch(config)#int f
Switch(config)#int fastEthernet 0/2
Switch(config-if)#swi
Switch(config-if)#switchport mode access
Switch(config-if)#sw
Switch(config-if)#switchport access vlan 3
Switch(config-if)#exit
Switch(config)#interface range f
Switch(config)#interface range fastEthernet 0/3-5
Switch(config-if-range)#swi
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#sw
Switch(config-if-range)#switchport ac
Switch(config-if-range)#switchport access vlan 2
Switch(config-if-range)#interface f
Switch(config-if-range)#exit
Switch(config)#int f
Switch(config)#int fastEthernet 0/1
Switch(config-if)#swi
Switch(config-if)#switchport mode trunk
Switch(config-if)#swi
Switch(config-if)#switchport mode trunk allowed vlan 2, 3
^
% Invalid input detected at '^' marker.

Switch(config-if)#switchport trunk allowed vlan 2, 3
^
% Invalid input detected at '^' marker.

Switch(config-if)#switchport trunk allowed vlan 2,3
Switch(config-if)#exit
```

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Top

1.2 Configure Inter-VLAN Routing on the Office Router

Enable the physical interface:

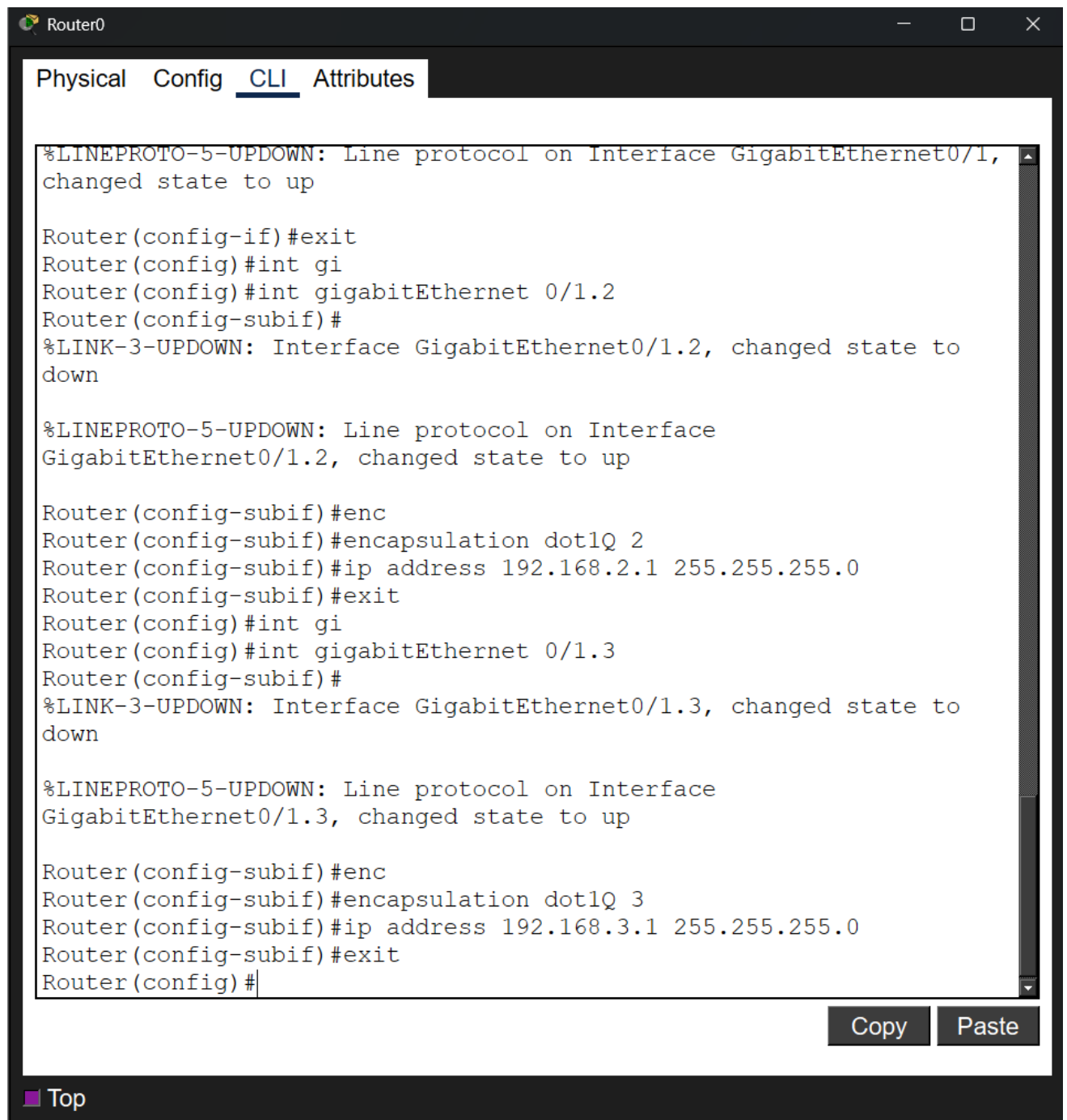
```
Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int g
Router(config)#int gigabitEthernet 0/1
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1,
changed state to up

Router(config-if)#exit
Router(config)#int gi
```

Create sub-interfaces for each VLAN:



The screenshot shows a Cisco Packet Tracer console window for Router0. The window has tabs for Physical, Config, CLI, and Attributes, with the CLI tab selected. The console output shows the following commands and status messages:

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1,
changed state to up

Router(config-if)#exit
Router(config)#int gi
Router(config)#int gigabitEthernet 0/1.2
Router(config-subif)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/1.2, changed state to
down

%LINEPROTO-5-UPDOWN: Line protocol on Interface
GigabitEthernet0/1.2, changed state to up

Router(config-subif)#enc
Router(config-subif)#encapsulation dot1Q 2
Router(config-subif)#ip address 192.168.2.1 255.255.255.0
Router(config-subif)#exit
Router(config)#int gi
Router(config)#int gigabitEthernet 0/1.3
Router(config-subif)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/1.3, changed state to
down

%LINEPROTO-5-UPDOWN: Line protocol on Interface
GigabitEthernet0/1.3, changed state to up

Router(config-subif)#enc
Router(config-subif)#encapsulation dot1Q 3
Router(config-subif)#ip address 192.168.3.1 255.255.255.0
Router(config-subif)#exit
Router(config)#
```

At the bottom right of the console window, there are buttons for "Copy" and "Paste". At the bottom left, there is a "Top" button.

1.3 Check the Connectivity

Assign IP:

The image displays four sequential screenshots of the Windows 'IP Configuration' window, each showing a different network interface configured with a static IP. In each window, the 'Static' radio button is selected, and the 'DHCP' button is disabled. The fields are filled with the following values:

- Window 1:** IPv4 Address: 192.168.2.2, Subnet Mask: 255.255.255.0, Default Gateway: 192.168.2.1, DNS Server: 0.0.0.0
- Window 2:** IPv4 Address: 192.168.2.3, Subnet Mask: 255.255.255.0, Default Gateway: 192.168.2.1, DNS Server: 0.0.0.0
- Window 3:** IPv4 Address: 192.168.2.4, Subnet Mask: 255.255.255.0, Default Gateway: 192.168.2.1, DNS Server: 0.0.0.0
- Window 4:** IPv4 Address: 192.168.3.2, Subnet Mask: 255.255.255.0, Default Gateway: 192.168.3.1, DNS Server: 0.0.0.0

Ping test

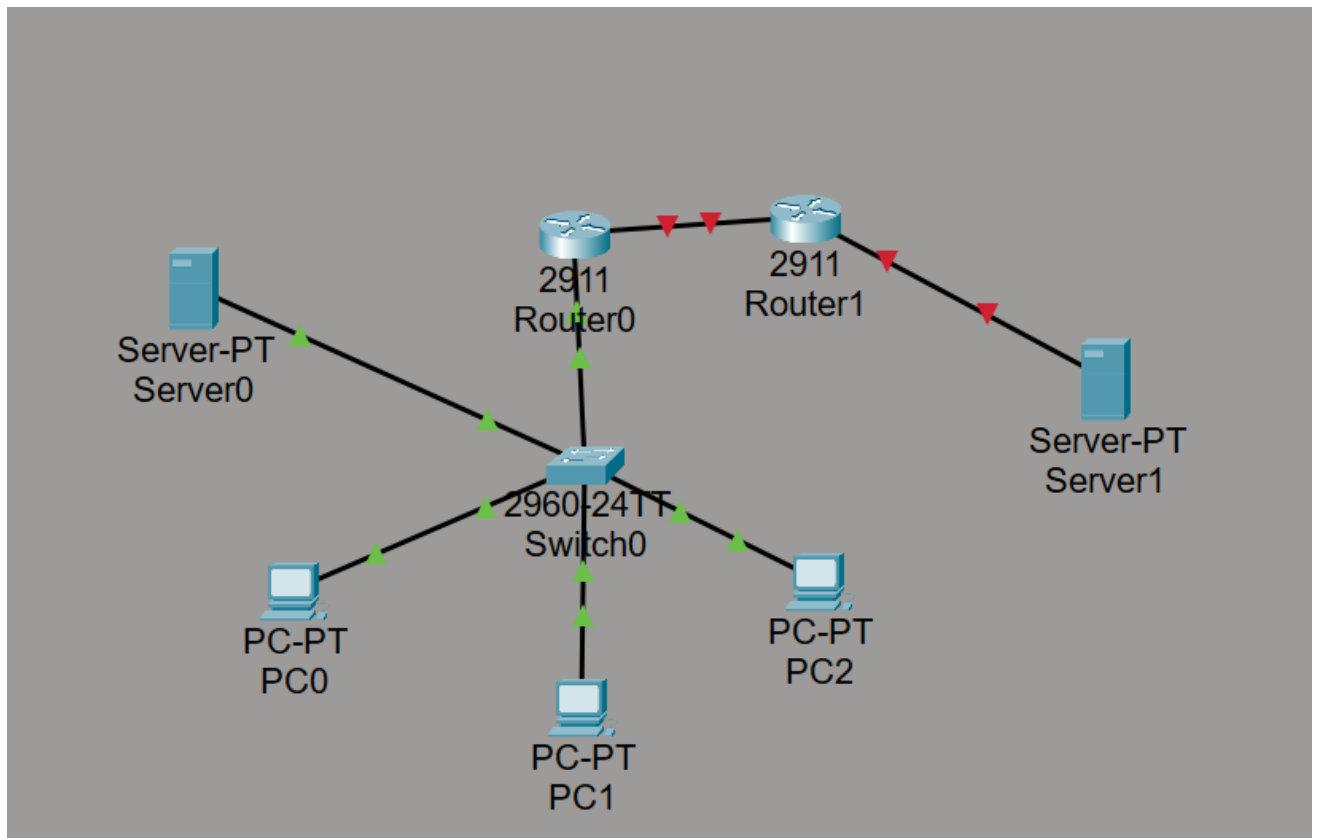
```
C:\>ping 192.168.3.2

Pinging 192.168.3.2 with 32 bytes of data:

Reply from 192.168.3.2: bytes=32 time<1ms TTL=127
Reply from 192.168.3.2: bytes=32 time<1ms TTL=127
Reply from 192.168.3.2: bytes=32 time<1ms TTL=127
Reply from 192.168.3.2: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.3.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Part 2: Internet Connectivity Setup



2.1 Configure ISP Router

Office router IP

```
Router(config)#int g0/0
Router(config-if)#ip address 213.234.10.1 255.255.255.252
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

Router(config-if)#exit
Router(config)#
```

ISP router IP

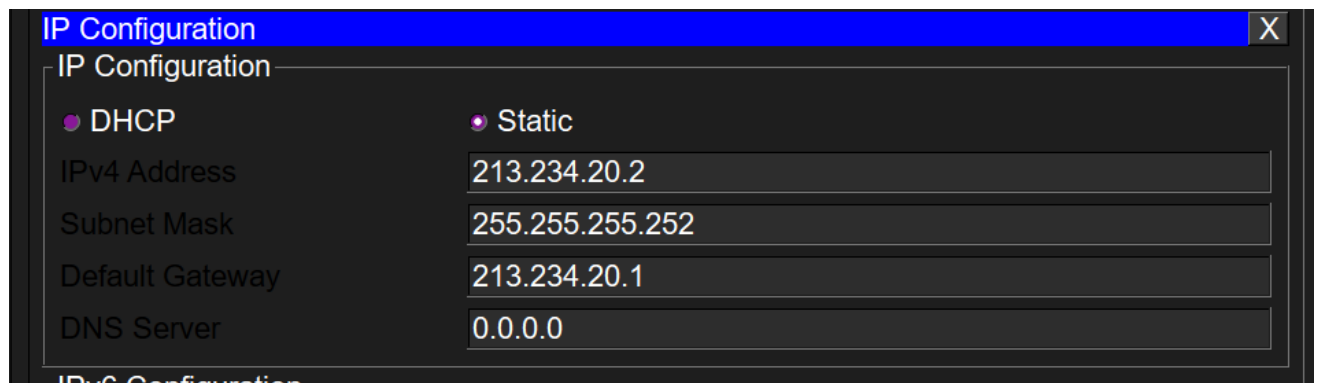
```
Router(config)#int g0/1
Router(config-if)#ip address 213.234.20.1
% Incomplete command.
Router(config-if)#ip address 213.234.20.1 255.255.255.252
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1,
changed state to up

Router(config-if)#exit
Router(config)#
```

Step 2: Set up IP address, subnet mask and gateway on Server1.



The screenshot shows a window titled "IP Configuration" with a close button (X) in the top right corner. Below the title bar, there is a section labeled "IP Configuration". Inside this section, there are two radio buttons: "DHCP" (which is unselected) and "Static" (which is selected). Below the radio buttons, there are four text input fields with the following values: "IPv4 Address" is "213.234.20.2", "Subnet Mask" is "255.255.255.252", "Default Gateway" is "213.234.20.1", and "DNS Server" is "0.0.0.0".

2.2 Configure Office Router for Internet Access

```
Router(config)#int g0/0
Router(config-if)#ip address 213.234.10.2 255.255.255.252
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#ip route 0.0.0.0 0.0.0.0 213.234.10.1
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#wr mem
Building configuration...
[OK]
Router#
```

Part 3: NAT Configuration

3.1 Dynamic NAT (PAT) for Internal Users on the Office Router

Define NAT inside/outside interfaces:

```
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int g0/0
Router(config-if)#ip nat outside
Router(config-if)#exit
Router(config)#int g0/1.2
Router(config-subif)#ip nat inside
Router(config-subif)#exit
Router(config)#int g0/1.3
Router(config-subif)#ip nat inside
Router(config-subif)#exit
Router(config)#
```

Create an Access Control List (ACL) for NAT traffic:

```
Router(config)#
Router(config)#ip access-list standard FOR-NAT
Router(config-std-nacl)#permit 192.168.2.0 0.0.0.255
Router(config-std-nacl)#permit 192.168.3.0 0.0.0.255
Router(config-std-nacl)#
```

Enable NAT Overload (PAT) on the WAN interface:

```
Router(config)#ip nat inside source list FOR-NAT interface
gigabitEthernet0/0 overload
Router(config)#
```

3.2 Static NAT for Web Server

```
Router(config)#ip nat inside source static tcp 192.168.3.2 80
213.234.10.2 80
Router(config)#
```

Part 4: Testing and Verification

4.1 Verify NAT Translations

```
Router#show ip nat translations
Pro  Inside global      Inside local      Outside local
Outside global
tcp  213.234.10.2:80     192.168.3.2:80   ---              ---
```

4.2 Verify End-to-End Connectivity

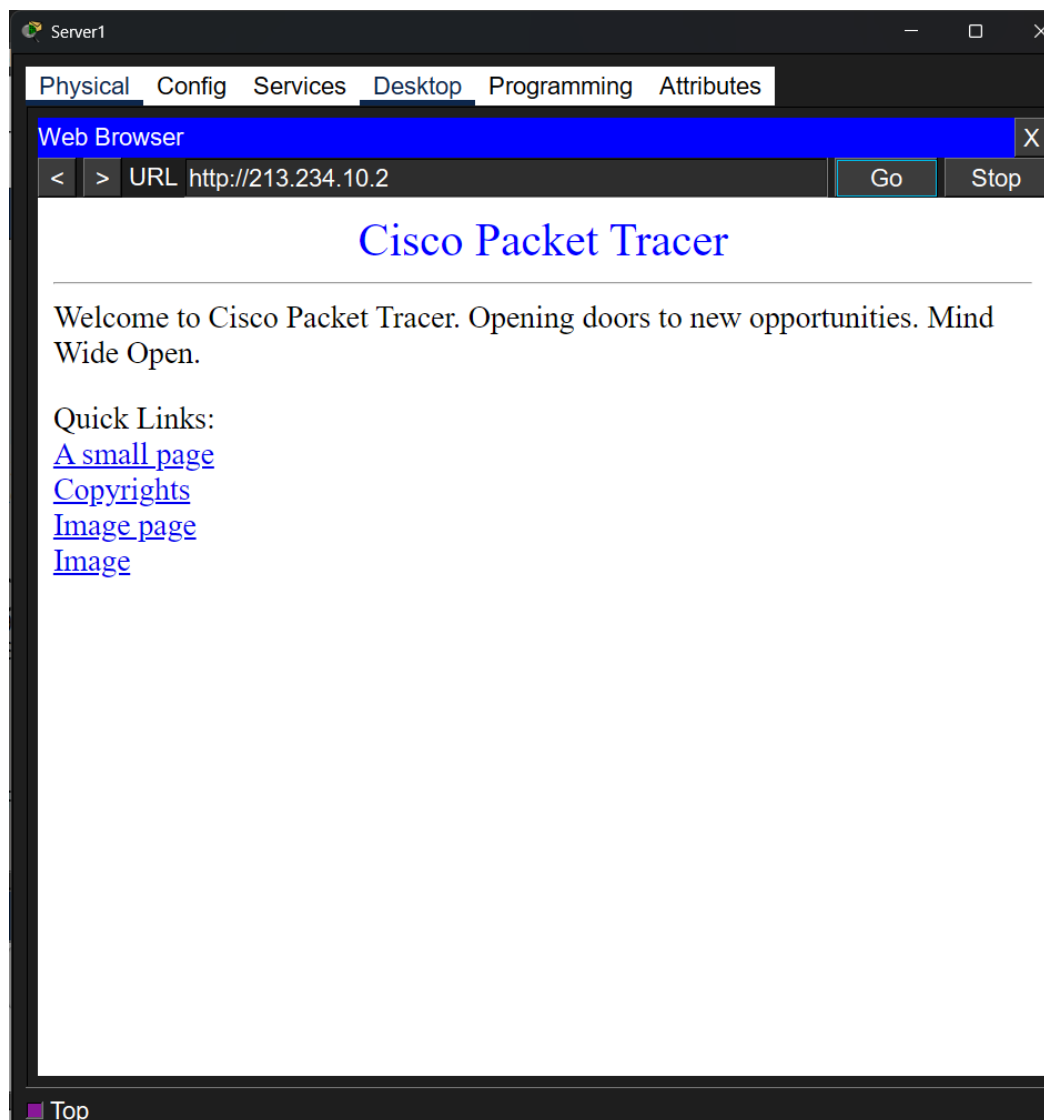
```
C:\>ping 213.234.20.2

Pinging 213.234.20.2 with 32 bytes of data:

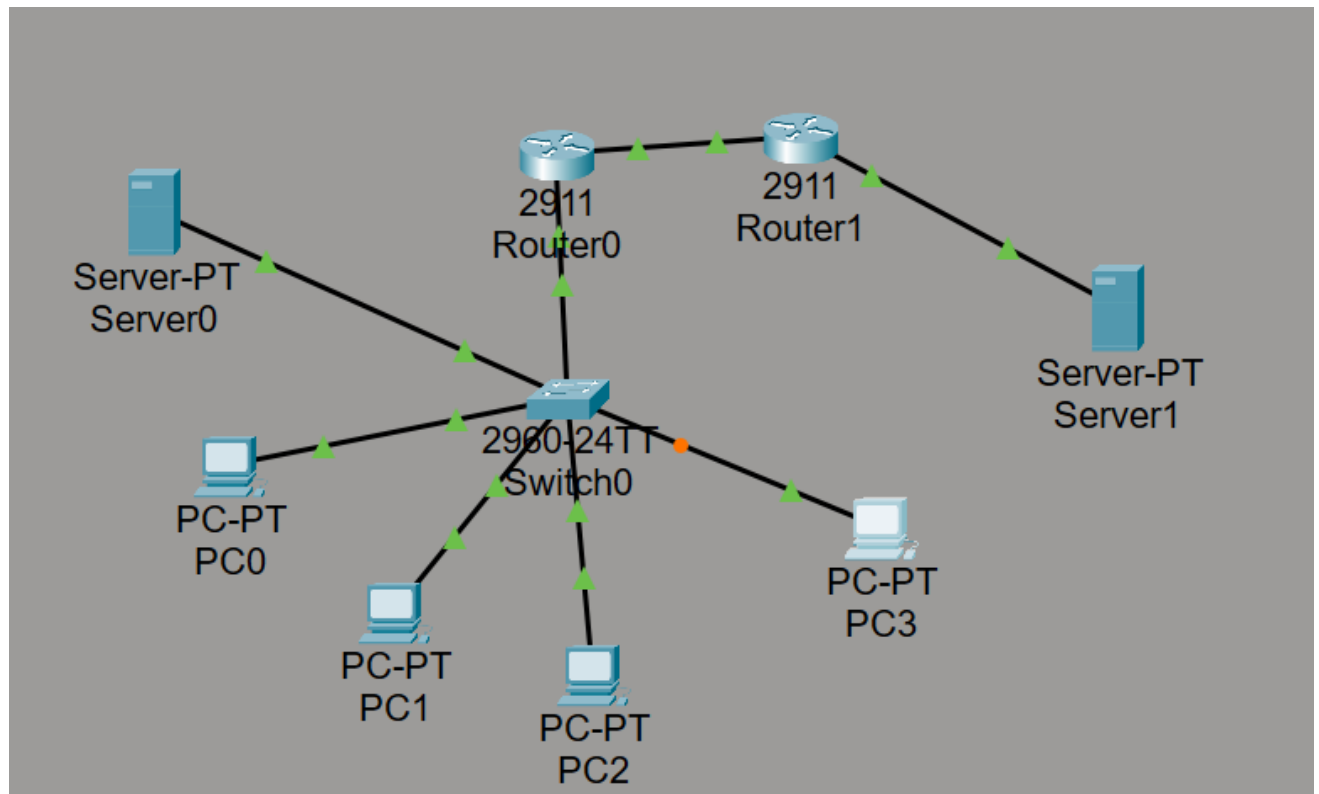
Reply from 213.234.20.2: bytes=32 time<1ms TTL=126
Reply from 213.234.20.2: bytes=32 time<1ms TTL=126
Reply from 213.234.20.2: bytes=32 time=1ms TTL=126
Reply from 213.234.20.2: bytes=32 time=4ms TTL=126

Ping statistics for 213.234.20.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 4ms, Average = 1ms

C:\>
```



Part 5: Independent Task



Create a new VLAN on the switch (e.g., VLAN 4) and name it Guests

```
Switch>
Switch>en
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 4
Switch(config-vlan)#name Guests
Switch(config-vlan)#exit
Switch(config)#int f
Switch(config)#int fastEthernet 0/6
Switch(config-if)#sw
Switch(config-if)#switchport mode acc
Switch(config-if)#switchport mode access
Switch(config-if)#sw
Switch(config-if)#switchport ac
Switch(config-if)#switchport access vlan 4
Switch(config-if)#exit
Switch(config)#
```

```
Switch(config)#int fastEthernet 0/1
Switch(config-if)#sw
Switch(config-if)#switchport mode tru
Switch(config-if)#switchport mode trunk
Switch(config-if)#sw
Switch(config-if)#switchport tr
Switch(config-if)#switchport trunk allowed vlan 2,3,4
Switch(config-if)#end
```

Configuring Inter-VLAN Routing on Router0

```
Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int g0/1.4
Router(config-subif)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/1.4, changed state to
down

%LINEPROTO-5-UPDOWN: Line protocol on Interface
GigabitEthernet0/1.4, changed state to up

Router(config-subif)#en
Router(config-subif)#encapsulation dot1Q4
^
% Invalid input detected at '^' marker.

Router(config-subif)#encapsulation dot1Q 4
Router(config-subif)#ip address 192.168.4.1 255.255.255.0
Router(config-subif)#ip nat inside
Router(config-subif)#exit
Router(config)#
```

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Set a static IP, subnet mask, and default gateway for the new PC.

IP Configuration	
☒ DHCP	☒ Static
IPv4 Address	192.168.4.2
Subnet Mask	255.255.255.0
Default Gateway	192.168.4.1
DNS Server	0.0.0.0

Update your NAT configuration to include the new subnet in the NAT ACL

```
Router(config)#ip access-list standard FOR-NAT
Router(config-std-nacl)#permit 192.168.4.0 0.0.0.255
Router(config-std-nacl)#exit
Router(config)#
```

Verify that the new PC can ping Server1 and access the web server using the public IP.

```
C:\>ping 213.234.20.2

Pinging 213.234.20.2 with 32 bytes of data:

Reply from 213.234.20.2: bytes=32 time<1ms TTL=126
Reply from 213.234.20.2: bytes=32 time<1ms TTL=126
Reply from 213.234.20.2: bytes=32 time<1ms TTL=126
Reply from 213.234.20.2: bytes=32 time=11ms TTL=126

Ping statistics for 213.234.20.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 11ms, Average = 2ms

C:\>
```



Part 6: Questions

1. What is the purpose of using subinterfaces in inter-VLAN routing?

They allow a single physical router port to be used to route traffic between multiple VLANs. Each subinterface acts as a separate gateway for its own virtual network.

2. Why do we need overload in PAT configuration?

It allows multiple devices on a local network to access the internet using just one public IP address.

3. What roles do inside and outside interfaces play in NAT?

This tells the router exactly where to replace addresses in packets.

Inside: Marks the interface facing the local network.

Outside: Marks the interface facing the external network.

4. How would NAT behave without a defined ACL?

NAT won't work. The ACL serves as a filter: it tells the router which subnets of traffic are allowed to forward. If there's no list or it's empty, the router won't understand that the addresses need to be changed.

5. Can static and dynamic NAT be used simultaneously? Explain.

Yes. For example, dynamic NAT is used to allow all employees to access the internet, while static NAT is used to allow the outside world to access a specific internal web server.